



Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-07

1. Find the interval of $f(x) = \frac{x^4}{12} - \frac{5x^3}{6} + 3x^2 + 7$. In which it is concave downward.
(1) (2, 3) (2) (2, -3)
(3) (-2, 2) (4) (-3, 3)
2. Find the interval of $f(x) = \frac{x^4}{12} - \frac{5x^3}{6} + 3x^2 + 7$. In which it is concave upward.
(1) $(-\infty, 0) \cup (3, \infty)$
(2) $(-\infty, 2) \cup (3, \infty)$
(3) $(-\infty, -1) \cup (1, \infty)$
(4) None of these
3. Find the points at which the function f given by $f(x) = (x-2)^4(x+1)^3$ has
(i) Local maxima
(ii) Local minima
(iii) Point of inflexion
4. The minimum value of $4e^{2x} + 9e^{-2x}$ is:
(1) 11 (2) 12
(3) 10 (4) 14
5. The number of point of inflection of $f(x) = \frac{\ln x}{x}$ is
(1) 0 (2) 1
(3) 2 (4) 3
6. Find the point of inflection of $f(x) = \frac{x^2-1}{x^2+1}$
7. If a right circular cone, having maximum volume, is inscribed in a sphere of radius 3 cm then the curved surface area (in cm^2) of this cone is:
(1) $6\sqrt{2}\pi$
(2) $6\sqrt{3}\pi$
(3) $8\sqrt{2}\pi$
(4) $8\sqrt{3}\pi$
8. Draw the graphs: $y = x^3 - 3x^2 + 2x$.
9. Draw the graphs: $y = (x+2)^2(x-1)$.
10. Draw the graphs: $y = \frac{1}{1+x^2}$.
11. Draw the graphs: $f(x) = x^2 + \frac{1}{x}$.
12. Draw the graphs: $f(x) = \frac{x}{x^2-1}$.
13. Find the set of value of m for the cubic $x^3 - \frac{3}{2}x^2 + \frac{5}{2} = \log_{1/4}(m)$ has 3 distinct solutions.

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

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|---------------------------------|---------------------------------------|
| 1. (1) | 7. (4) |
| 2. (2) | 8. See the solution |
| 3. (i) $x = 2/7$; | 9. See the solution |
| (ii) $x = 2$; | 10. See the solution |
| (iii) $x = -1$ | 11. See the solution |
| 4. (2) | 12. See the solution |
| 5. (2) | 13. $\frac{1}{32} < m < \frac{1}{16}$ |
| 6. $x = \pm \frac{1}{\sqrt{3}}$ | |



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>